

Chapter 56 — Hypospadias — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Commonest complication:** urethrocutaneous fistula (just under 10%).
- **Fistula repair:** wait ≥ 6 –12 months; exclude distal obstruction first.
- **Meatal stenosis:** symptoms + calibration < 8 Fr prepubertal or < 12 Fr postpubertal.
- **Recurrent curvature or torsion:** $> 30^\circ$.
- **Buccal mucosa** = graft of choice; **lip within the glans**.
- **Severe proximal hypospadias** $\rightarrow$ staged: curvature first, urethroplasty second.
- **25–75%** need extra procedures after planned two-stage repair.
- **Distal success 83–95%**; proximal complications **23–68%**.
- **BXO** (balanitis xerotica obliterans): excise all, staged buccal; recurrence up to 40%.
- **Follow up through puberty:** only about half of complications appear in year 1.

PROXIMAL REPAIR AND POSTOPERATIVE CARE

- **Poiseuille:** resistance $\propto$ length, $\propto 1/\text{radius}^4 \rightarrow$ long neourethra = stricture risk.
- **Glans non-distensible** + no spongiosum $\rightarrow$ **diverticulum**.
- **TIP** (tubularised incised plate) long neourethra $\rightarrow$ flattened flow; **onlay** more elastic.
- **Modified Ulaanbaatar:** Duckett tube + Byars flaps; second stage at 6 months.
- **Diversion 3–21 days;** soft stent in complex repairs $\rightarrow$ less stenosis and fistula.
- **Oxybutynin** until stent removal (bladder spasm).
- **Foley** must be fixed in the midline; lateral pull $\rightarrow$ **glans dehiscence**.
- **Ibuprofen and ketorolac:** no added bleeding; paracetamol + ibuprofen regular for 48–72 h.

COMPLICATIONS

Definitions (Table 56.1):

- **Glans dehiscence:** fusion < 2 mm or coronal or more proximal meatus.
- **Skin surplus** ≥ 2 cm; **torsion** $> 30^\circ$.
- **Non-retractable reconstructed prepuce:** acceptable before puberty, abnormal after.

Fistula:

- Commonest at the **original meatus or coronal margin**.
- **Small early fistula rarely closes** spontaneously.
- **Work-up:** bougie à boule $\rightarrow$ urethroscopy $\rightarrow$ saline or methylene blue stress test.
- **Small shaft** $\rightarrow$ excise tract, multilayer offset closure.
- **Coronal fistula** $\rightarrow$ revise the urethroplasty (implies distal narrowing).
- **Recurrence about 20%;** reduce with tunica vaginalis or dartos layers.
- Diversion: none if simple; **7–14 days** if complex.

Meatus and glans:

- **Glans dehiscence:** reoperate only if symptomatic; inlay graft.
- **Postoperative meatus ≥ 8 Fr;** "laser-beam" stream.
- **Routine dilatation unnecessary;** refractory $\rightarrow$ dorsal meatotomy or redo.

Stricture:

- Sites: **native–neourethral junction**, coronal margin, glans.
- **Characterise by cystoscopy**.
- **Urethrotomy or dilatation rarely durable**, even < 1 cm.
- Refractory $\rightarrow$ **revision urethroplasty** (flap or buccal; often staged).
- **Hair + stones + UTI** (urinary tract infection) $\rightarrow$ resect and replace.
- **Enlarged utricle** with recurrent UTI $\rightarrow$ excise or fulgurate.

Diverticulum and curvature:

- **Diverticulum:** commoner after preputial flap, two-stage, proximal repairs; excise + multilayer.
- **BXO:** topical betamethasone or clobetasol, tacrolimus $\rightarrow$ usually fails $\rightarrow$ cutback, staged buccal.
- **BXO invades buccal mucosa**.
- **Curvature check:** deglove + artificial erection; **tourniquet hides basal curvature**.
- **Buried penis:** anchor Buck's fascia to dermis; scrotoplasty for penoscrotal transposition.

HYOSPADIAS OUTCOMES

- **Pfistermuller TIP meta-analysis:** complications 10.6%, reoperation 4.5%.
- TIP complications: **fistula 5.7% > meatal stenosis 3.6% > stricture 1.3%**.
- **Extra dartos layer** $\rightarrow$ fewer complications.
- **TIP vs Mathieu:** equal fistula; **more meatal stenosis with TIP**.
- **Recurrent UTI about 2%** $\rightarrow$ anatomical imaging (narrowing, VUR, diverticulum, utricle).
- **TIP $\rightarrow$ raised PVR** (post-void residual); **OIF** (onlay island flap) $\rightarrow$ diverticulum or fistula.
- **Fistula and dehiscence early; stricture late**.
- **Proximal fistula 3–45%;** recurrent curvature after two-stage repair **0–10%**.
- **STAG** (staged tubularised autograft) $< 25\%$; **Byars two-stage 30–70%** (43–81% across centres).
- **Proximal TIP:** no longer recommended (recurrent curvature).
- **Staged repair** $\rightarrow$ longer penis, better cosmesis.
- **Glans width < 14 mm,** proximal meatus, no barrier flap, previous surgery = risk.
- **Instruments:** HOSE (Hypospadias Objective Scoring Evaluation); PPS (Penile Perception Score: patient and parents); GPS (Genital Perception Score); HOPE (Hypospadias Objective Penile Evaluation: independent panels, photographs).
- **HOPE:** 1–10 per item; torsion and curvature 0 – $30^\circ = 10$, $> 70^\circ = 1$.
- **HRQoL** (health-related quality of life) domains: appearance, sexual, psychological, social, voiding.
- **Mild undiagnosed distal** $\rightarrow$ minimal sexual, fertility or voiding problems.

REOPERATION

- **Devine:** avoid "small operations"; resect scar, major tissue shifts.
- **Always artificial erection;** low threshold for cystoscopy.
- **SOS** (split-onlay skin) flap: about 1 cm pedicled skin, split for onlay + cover.
- **Nitroglycerin** topical for 24 h $\rightarrow$ flap flow (no RCT, randomised controlled trial).

- **Tissue expander:** skin sufficient 95%; complications 41%.
- **Bladder mucosa:** prolapse up to 38%; superseded.
- **Buccal:** harvest **10% oversize**; fenestrate 18–21 G; quilting sutures.
- **Wait ≥6 months** (to 12) before tubularising the graft.
- **Graft massage** from 2 weeks; topical steroid from 1 month.
- **Staged buccal > single-stage; dorsal inlay** best in redo.
- **Buccal staged success 30–75%**; meatal stenosis commonest failure.
- **Cecil-Culp:** penis buried in scrotum; release at 6–8 weeks (Cecil) or **9–12 months** (modified).
- **Donor site:** leave open (heals by secondary intention).

CLASSIC TRAPS

- **Trap:** small early fistula will close — **Actually:** rarely; repair at 6–12 months.

- **Trap:** excise a coronal fistula simply — **Actually:** revise the urethroplasty.
- **Trap:** repair a fistula without checking distally — **Actually:** distal obstruction causes recurrence.
- **Trap:** calibration <8 Fr alone = stenosis — **Actually:** symptoms are also required.
- **Trap:** routine meatal dilatation — **Actually:** unnecessary.
- **Trap:** assess curvature with a tourniquet — **Actually:** it hides basal curvature.
- **Trap:** meatotomy for BXO stenosis — **Actually:** fails; excise and stage.
- **Trap:** cheek graft for the glans — **Actually:** lip (thinner, fewer dehiscences).
- **Trap:** most complications appear in year 1 — **Actually:** about half; strictures late.
- **Trap:** small redo procedures after failure — **Actually:** Devine; radical reconstruction.